Learnings & Reflections Document

1. Team Reflection

2. **How did the team work together?**
   Initially, there were some differences in opinions and a bit of conflict between us, especially while deciding how to approach the build. But as the project progressed, we started dividing tasks more clearly and focusing on getting things done instead of arguing. Eventually, the workflow became smoother and more collaborative.
3. **What worked well?**
   Once roles became clearer, we were able to work more efficiently. Constant testing and small discussions helped us improve the robot step by step.
4. **What could be improved?**
   Better communication from the beginning would have saved time. Some confusion could have been avoided if we had planned roles and responsibilities earlier.

2. Technical Reflection

Key technical challenges

One of the main challenges was getting all the servo motors to move in coordination. Even small mistakes in wiring caused unexpected issues, which showed how important basic connections are. Power stability was also something we had to manage carefully.

What did you learn (hardware/software)?

We learned that small details like proper wiring and connections are extremely important and can affect the whole system. Servo calibration turned out to be a major factor—slight differences in angles could completely change the walking motion. We also understood how I2C communication and PWM signals work together to control multiple motors.

Mistakes & debugging experiences

Initially, we assumed bigger components or code issues were causing problems, but many times the issue was just loose wires or incorrect connections. This taught us to check simple things first before assuming major faults.

We also faced an issue where our PCA9685 module was not working properly, which initially confused us as we thought the problem was in our wiring or code. After testing and troubleshooting, we realized the component itself was faulty and had to get it replaced.

3. Design Reflection

Did the experience match your intentions?
To some extent, yes. The robot was able to walk and demonstrate coordinated motion, which was our main goal. However, the movement was not as smooth or natural as we initially imagined.

What would you improve?
We would improve the overall stability and smoothness of movement. Also, choosing lighter and more suitable components would help reduce strain on the motors and improve performance.

User feedback insights
Users found the robot interesting, but some pointed out that the movement felt slightly mechanical. This made us realize that not just functionality, but also the aesthetic and feel of motion matters in such projects.

4. Personal Reflection

What I learned
I learned how important small details are in physical projects, especially wiring and calibration. I also understood how different components work together as a system.

What I struggled with
I struggled with getting the movement right and debugging issues that didn't have obvious causes. Coordination between parts took a lot of trial and error.

What I am proud of
I am proud that we were able to make the robot walk/ crawl and improve it over time. Also, despite initial disagreements, we managed to complete the project properly as a team.